

ECE 3317

Applied Electromagnetic Waves

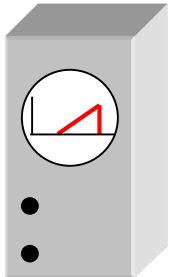
Prof. David R. Jackson

Fall 2025

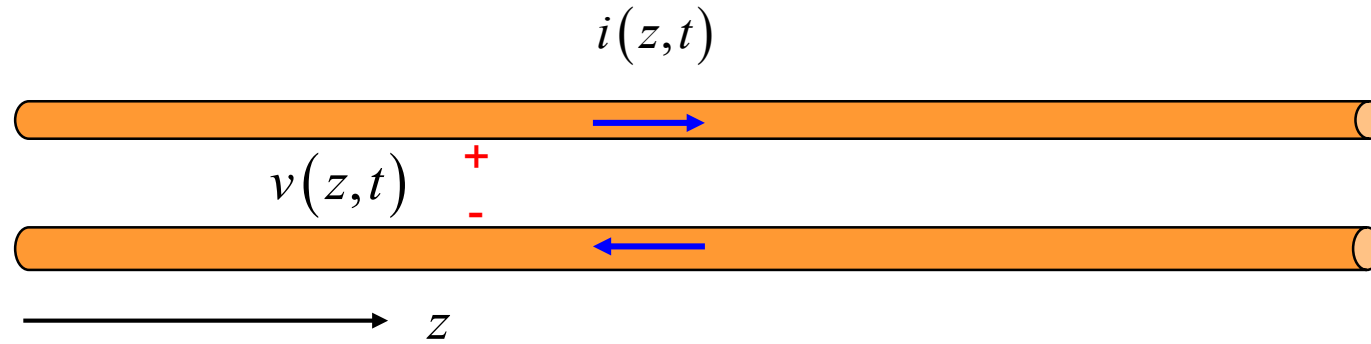
Notes 7

Transmission Lines

(Pulse Propagation and Reflection)



Pulse on Transmission Line (cont.)



Convenient for visualizing a wave traveling on the line (a “snapshot”).

$$v(z, t) = F(z - c_d t) + G(z + c_d t)$$

Alternative notation:

$$\uparrow \\ v^+(z, t)$$

$$\uparrow \\ v^-(z, t)$$

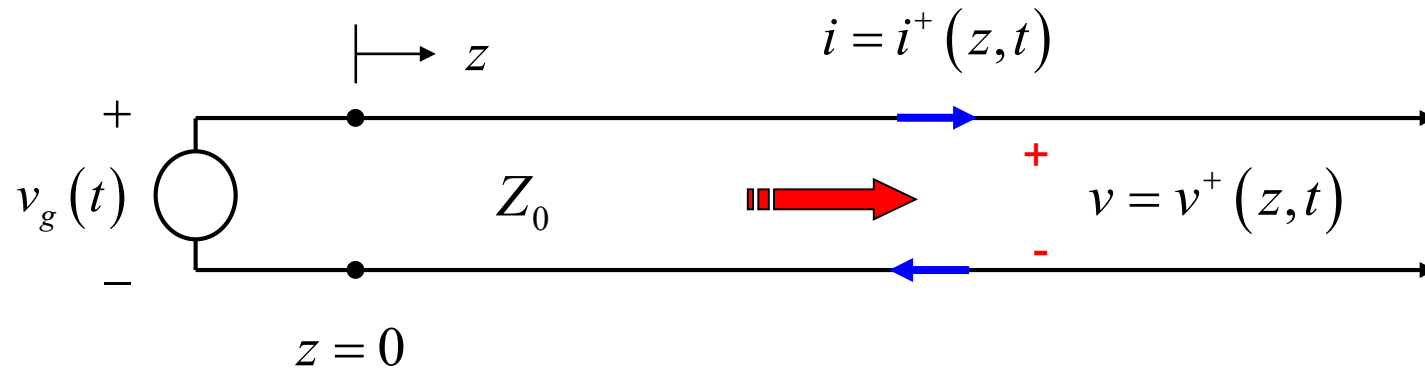
$$v(z, t) = f(t - z / c_d) + g(t + z / c_d)$$

Convenient for seeing what an oscilloscope on the line will display.

(In this set of notes we will be using the second form.)

Pulse on Transmission Line

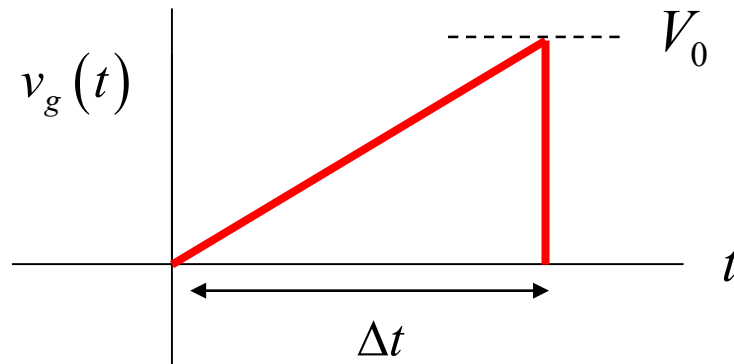
A voltage signal is applied at the input of a semi-infinite transmission line.



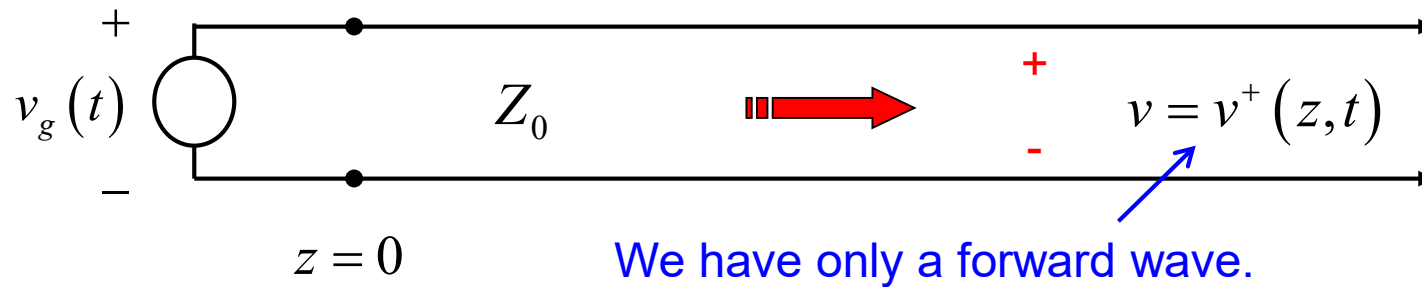
$$\frac{v^+(z, t)}{i^+(z, t)} = Z_0$$

Sawtooth wave voltage pulse

Example:



Pulse on Transmission Line (cont.)



$$v^+(z, t) = f(t - z / c_d)$$

Set $z = 0$: $v^+(0, t) = f(t)$

What is the function f ?

Also, we have: $v^+(0, t) = v_g(t)$

$\Rightarrow f(t) = v_g(t)$

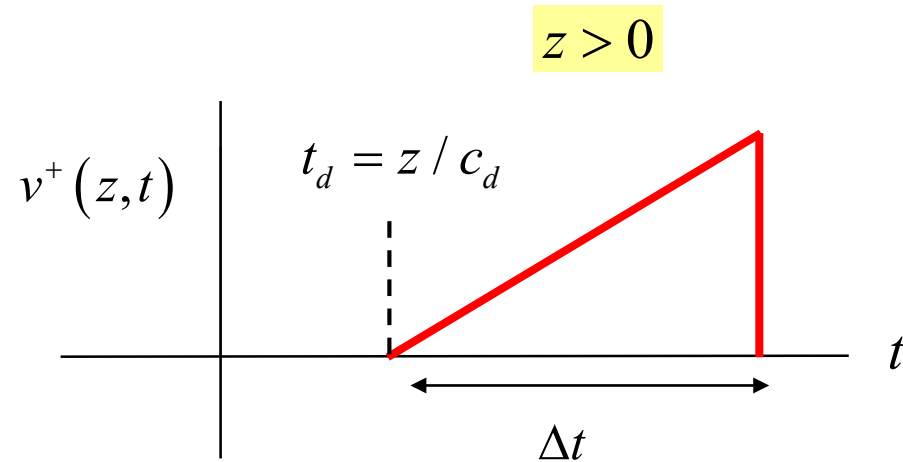
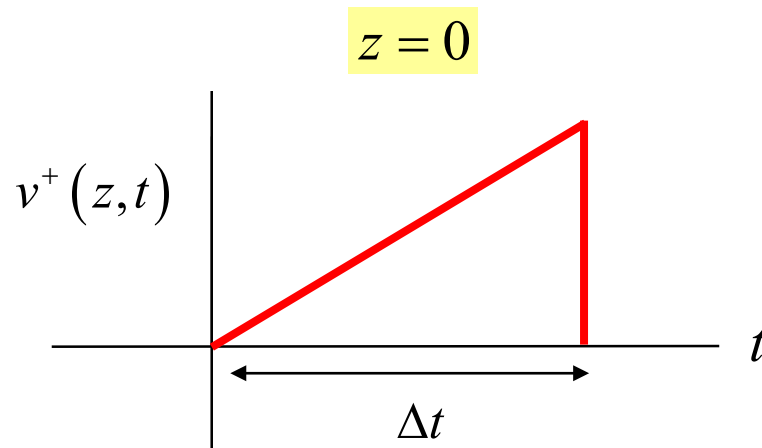
Hence

$$v^+(z, t) = v_g(t - z / c_d)$$

Pulse on Transmission Line (cont.)

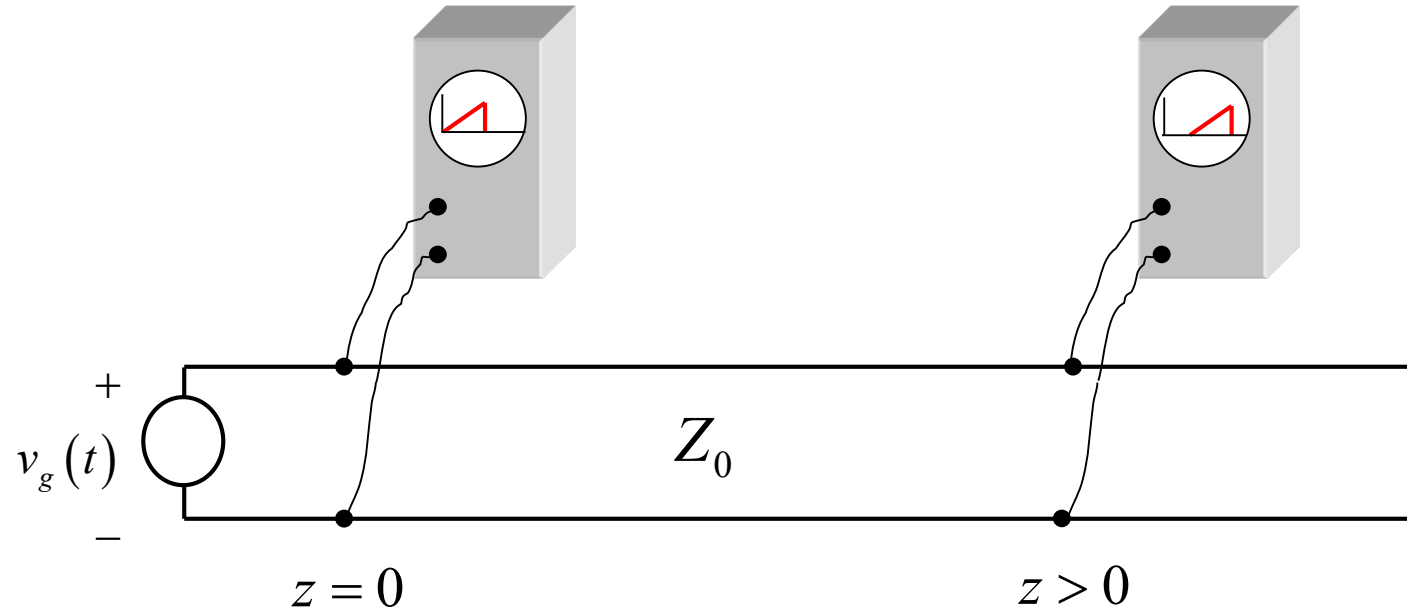
$$v^+(z, t) = v_g(t - z / c_d)$$

At any position z , the voltage pulse (oscilloscope trace) that is measured is the same as the input voltage pulse from the generator, except that it is delayed by a time $t_d = z / c_d$.



Pulse on Transmission Line (cont.)

Note the delay in the trace on this oscilloscope.



Time delay:

$$t_d = z / c_d$$

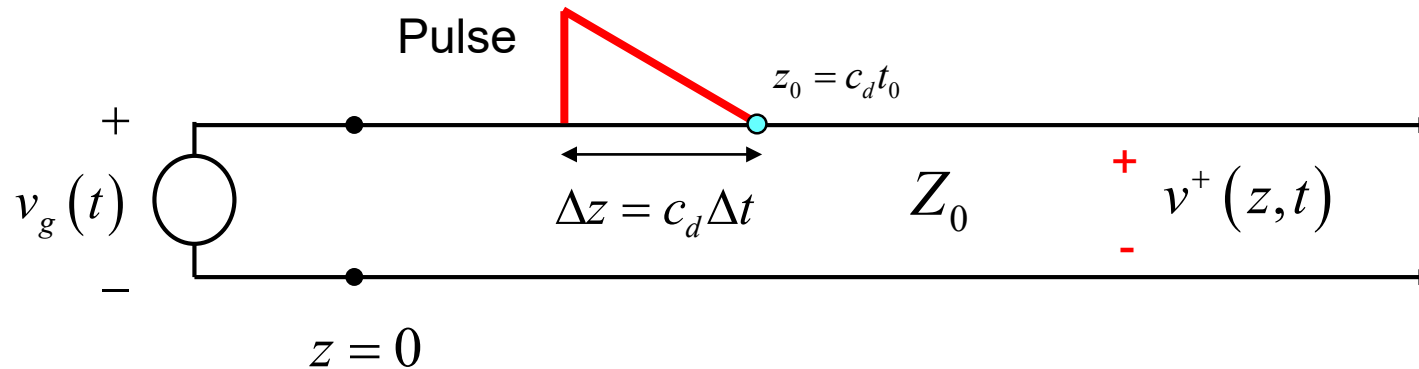
Here is what oscilloscopes will show.

$$v^+(z, t) = v_g(t - z / c_d)$$

Pulse on Transmission Line (cont.)

“Snapshot” of voltage wave at a fixed time.

$$t = t_0$$



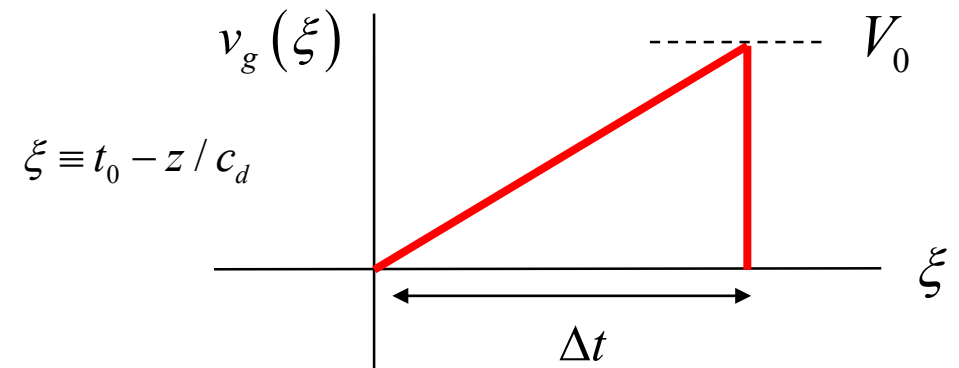
Note:

The “tip” of the sawtooth gets launched first on the line.

Note that the shape of the pulse as a function of z is a scaled mirror image of the pulse shape as a function of t .

$$v^+(z, t_0) = v_g(t_0 - z / c_d)$$

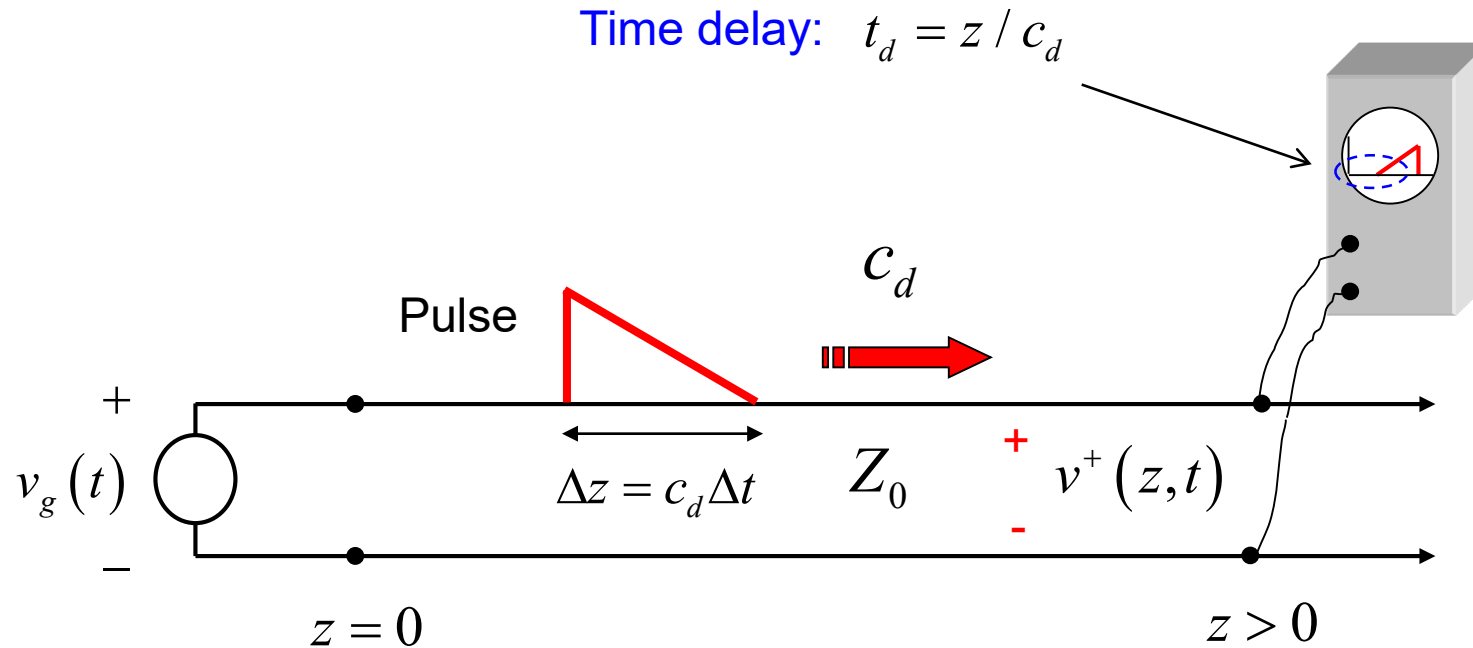
Note the minus sign! This causes the mirror-image effect.



Pulse on Transmission Line (cont.)

This voltage waveform moves past the oscilloscope to create the trace.

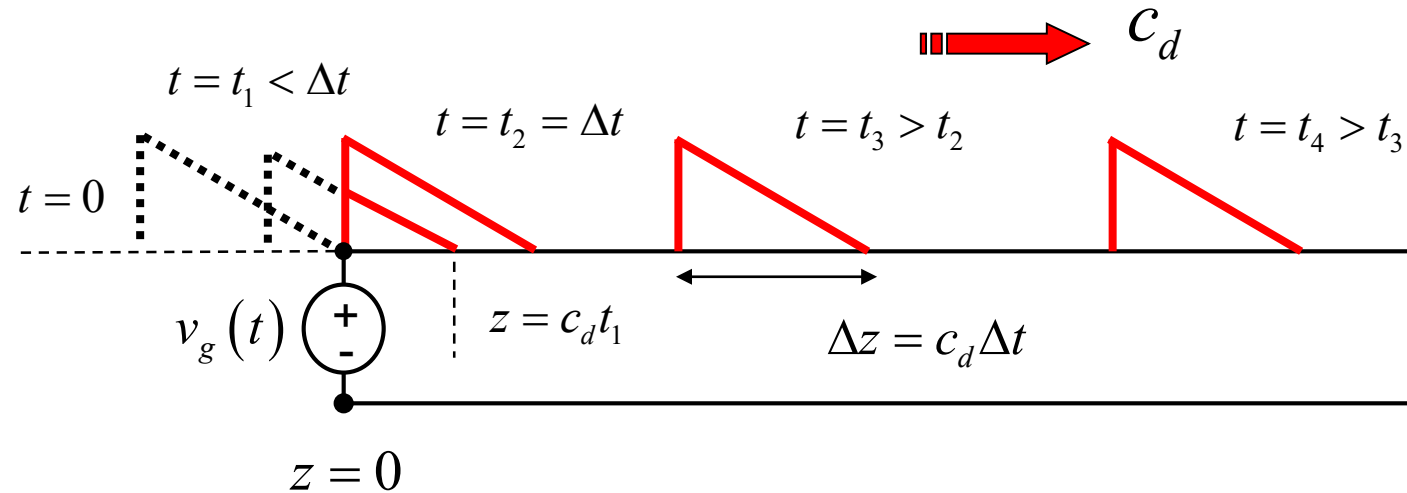
(The oscilloscope trace is the mirror image of the snapshot shape.)



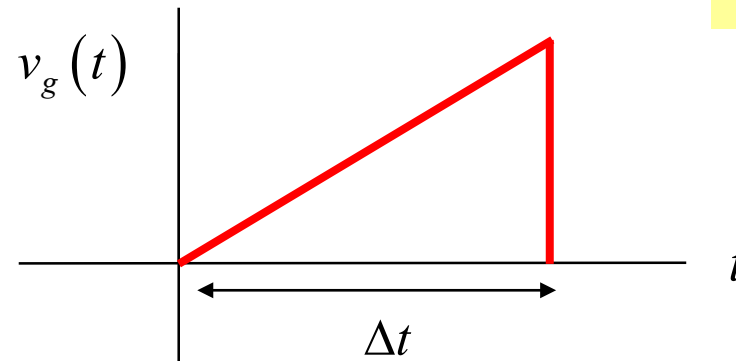
Pulse on Transmission Line (cont.)

The pulse is shown emerging from the source end of the line.

A series of “snapshots” is shown.

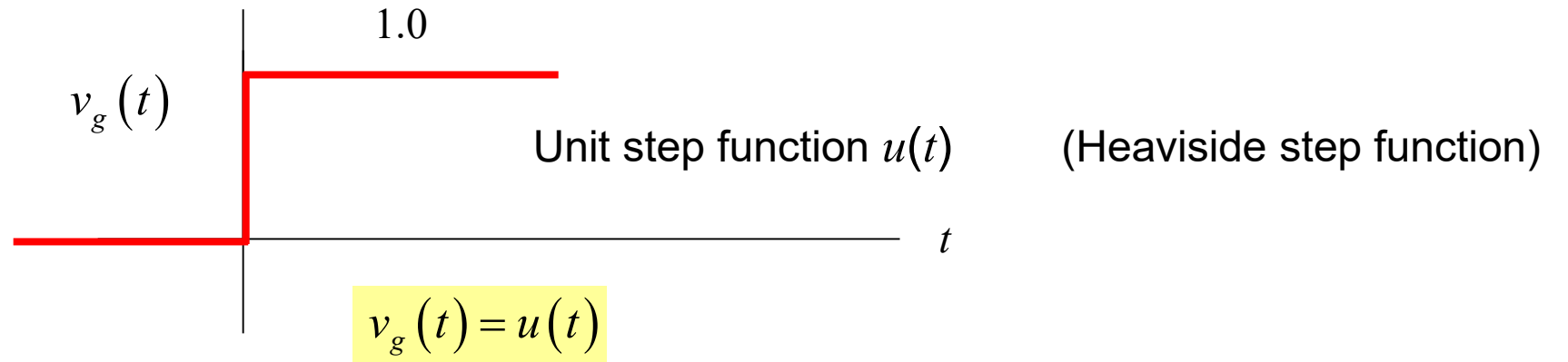
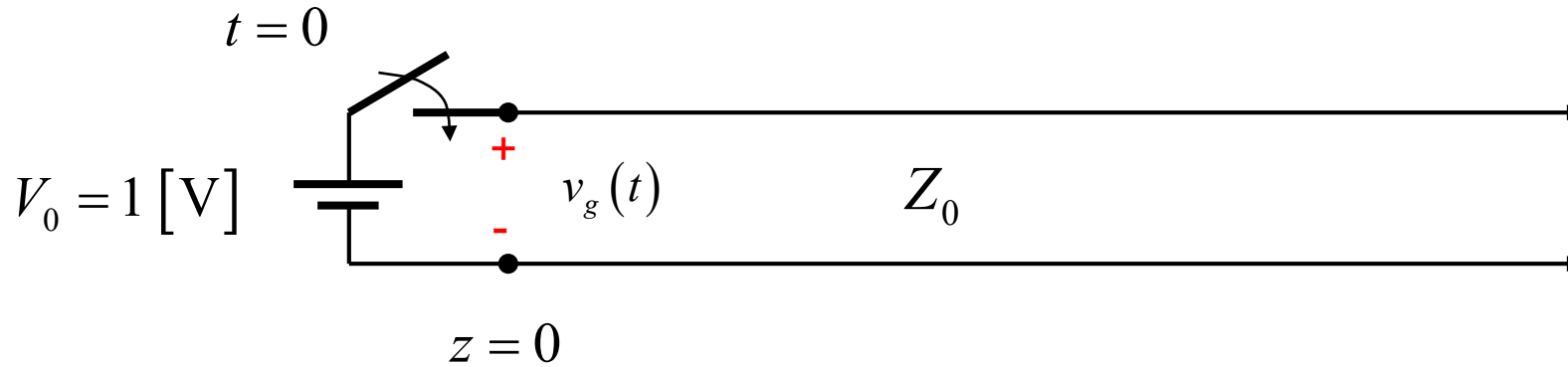


$$v^+(z, t) = v_g(t - z / c_d)$$



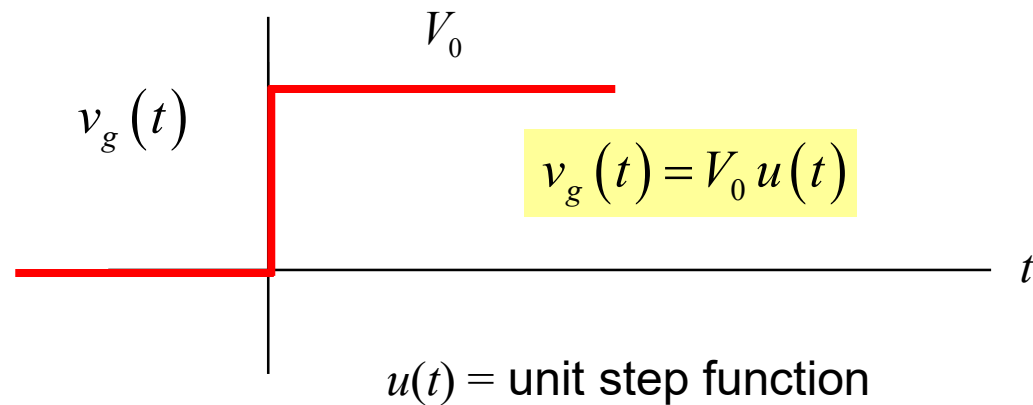
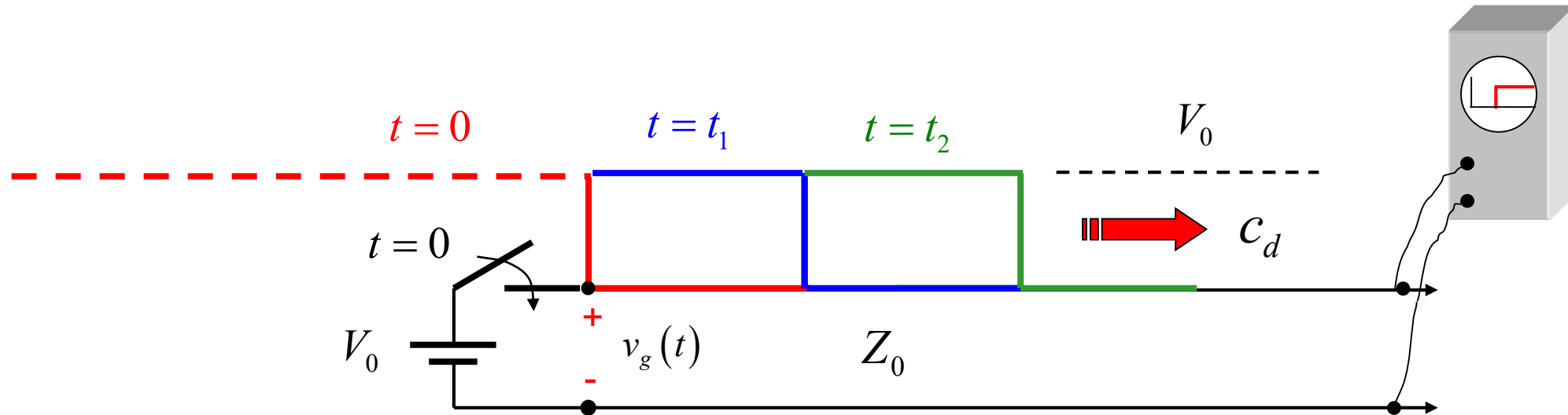
Step Function Source

Another example (battery and switch)



Step Function Source (cont.)

The step function is shown propagating down the line.



$$v^+(z, t) = V_0 u(t - z / c_d)$$

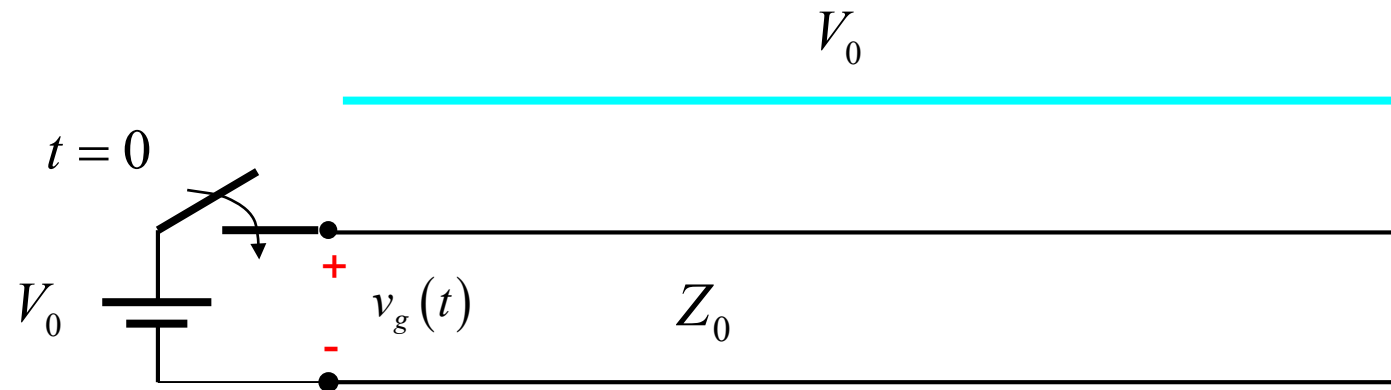
$$v_g(t) = V_0 u(t)$$

$u(t)$ = unit step function

Step Function Source (cont.)

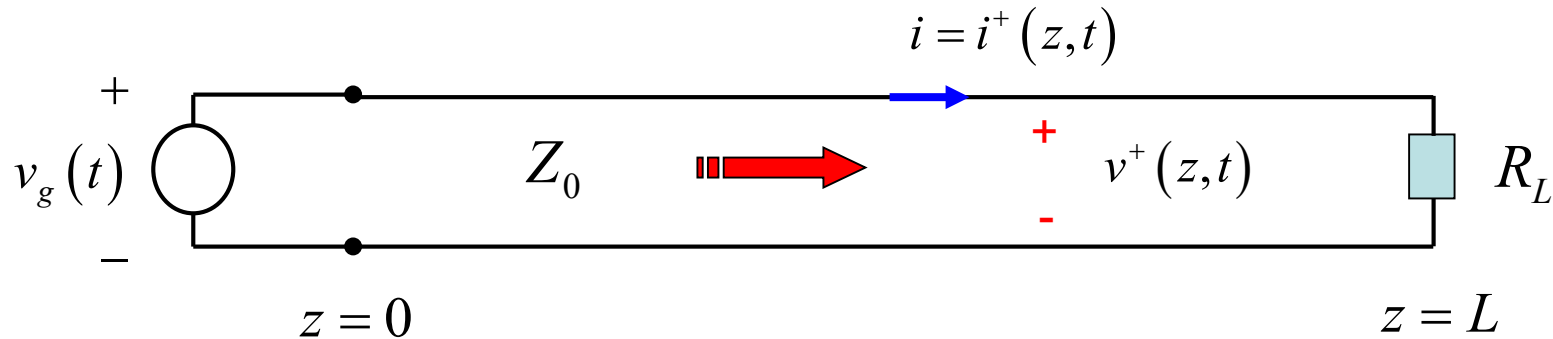
Steady-state solution ($t = \infty$)

The steady-state voltage is just the battery voltage.



Matched Load

$$R_L = Z_0$$



At the load:

Incident wave: $\frac{v^+(L,t)}{i^+(L,t)} = Z_0$

Total voltage: $\frac{v(L,t)}{i(L,t)} = R_L$

same

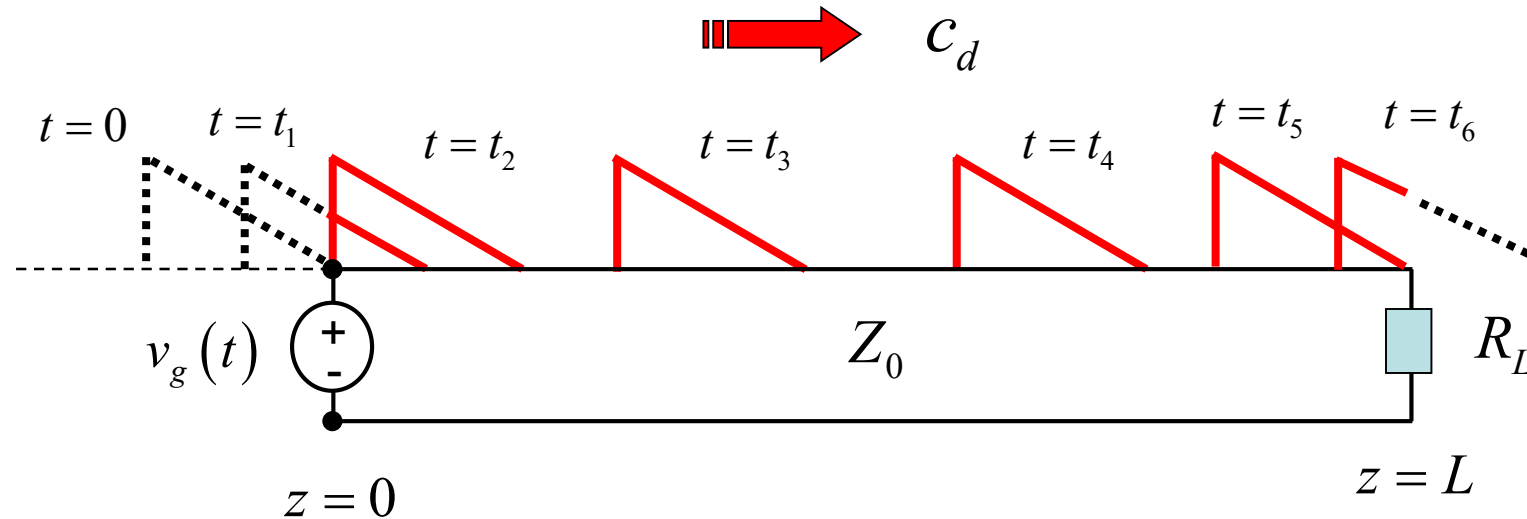
$$v^+(L,t) = v(L,t)$$

Recall: $Z_0 \equiv \frac{v^+(z,t)}{i^+(z,t)}$

At $z = L$, the two right-hand side terms are the same since $R_L = Z_0$. Hence the total voltage at the load is the same as the incident voltage. There is thus **no reflection**. When the waveform hits the load, it “sees” a continuation of the line.

Absorption by Load

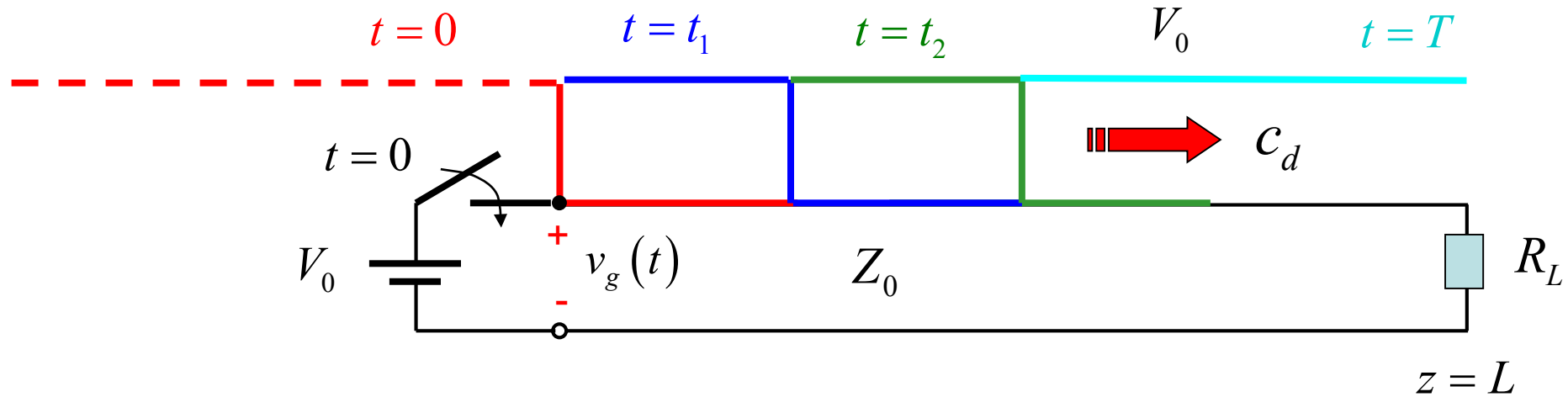
This shows the sawtooth waveform propagating on a matched line.



The pulse is shown emerging from the source end of the line, traveling down the line, and then being absorbed by the matched load.

Absorption by Load (cont.)

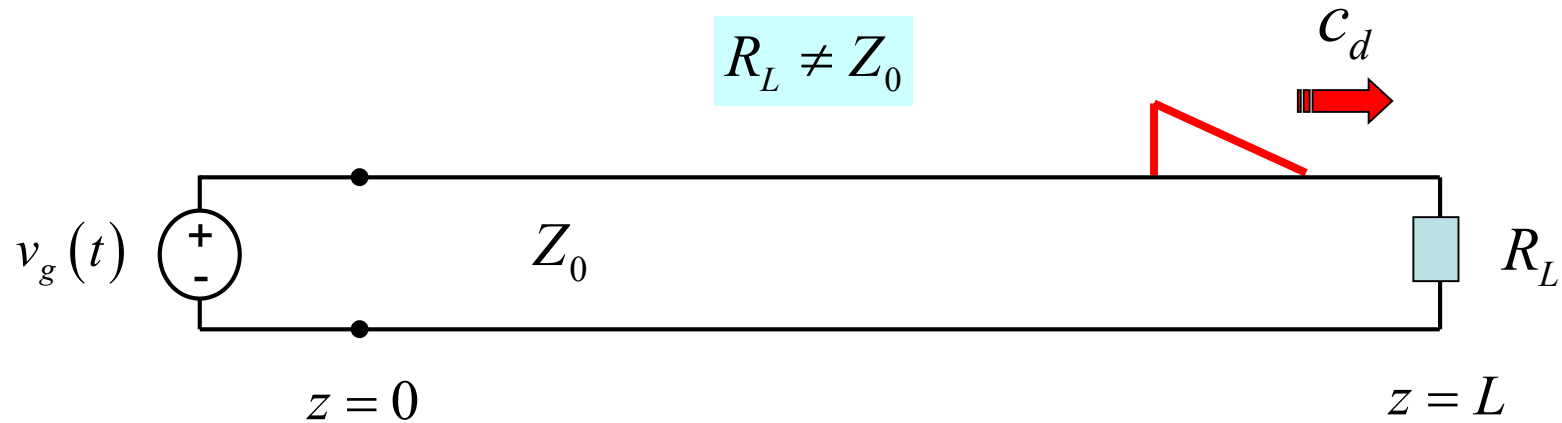
This shows the step function waveform propagating on a matched line.



Time to reach the load end: $T = \frac{L}{c_d}$

For $t > T$ we have reached steady state: $V(z, t) = V_0$ everywhere on the line.

Load Reflection



On the line:

$$v(z, t) = f(t - z / c_d) + g(t + z / c_d)$$

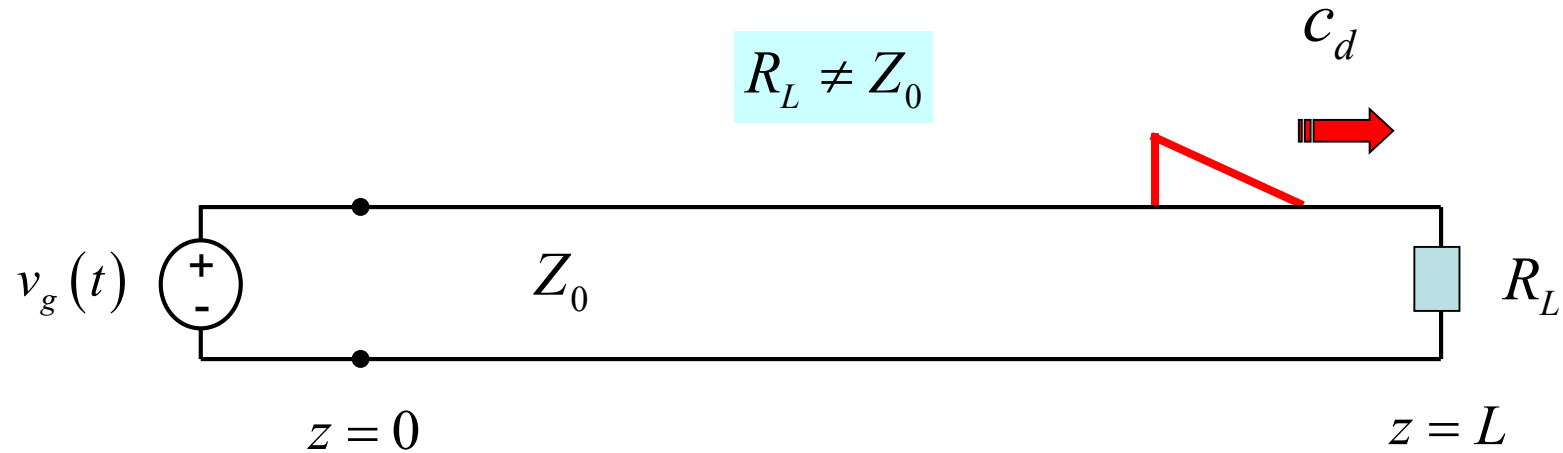
$$i(z, t) = \frac{1}{Z_0} [f(t - z / c_d) - g(t + z / c_d)]$$

where $v^+(z, t) = f(t - z / c_d) = v_g(t - z / c_d)$ (known function)

Goal: solve for $v^-(z, t) = g(t + z / c_d)$

At the load we know this: $\frac{v(L, t)}{i(L, t)} = R_L$

Load Reflection (cont.)



At the load:

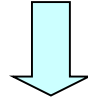
$$\frac{v(L,t)}{i(L,t)} = R_L$$

Hence, we have:

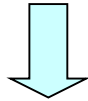
$$\frac{\begin{matrix} v^+(L,t) & v^-(L,t) \\ \downarrow & \downarrow \\ f(t-L/c_d) + g(t+L/c_d) \end{matrix}}{\frac{1}{Z_0} [f(t-L/c_d) - g(t+L/c_d)]} = R_L$$

Load Reflection (cont.)

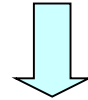
$$f(t - L / c_d) + g(t + L / c_d) = R_L \left(\frac{1}{Z_0} [f(t - L / c_d) - g(t + L / c_d)] \right)$$



$$g(t + L / c_d) \left[1 + \frac{R_L}{Z_0} \right] = f(t - L / c_d) \left[-1 + \frac{R_L}{Z_0} \right]$$



$$g(t + L / c_d) = f(t - L / c_d) \left(\frac{-1 + \frac{R_L}{Z_0}}{1 + \frac{R_L}{Z_0}} \right)$$



$$g(t + L / c_d) = f(t - L / c_d) \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

Load Reflection (cont.)

$$g(t + L / c_d) = f(t - L / c_d) \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

Relabel for simplicity:

$$v_L^+(t) \equiv f(t - L / c_d) \quad \text{voltage of forward-traveling waveform at the load}$$

$$v_L^-(t) \equiv g(t + L / c_d) \quad \text{voltage of backward-traveling waveform at the load}$$

Then

$$v_L^-(t) = v_L^+(t) \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

Define

$$\Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right) \quad \text{load reflection coefficient}$$

We then have

$$v_L^-(t) = \Gamma_L v_L^+(t)$$

Load Reflection (cont.)

Summary for load reflection

$$v_L^-(t) = \Gamma_L v_L^+(t)$$

$$\Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

load reflection coefficient

Note:

There is no reflection for a matched system ($R_L = Z_0$).

Note:

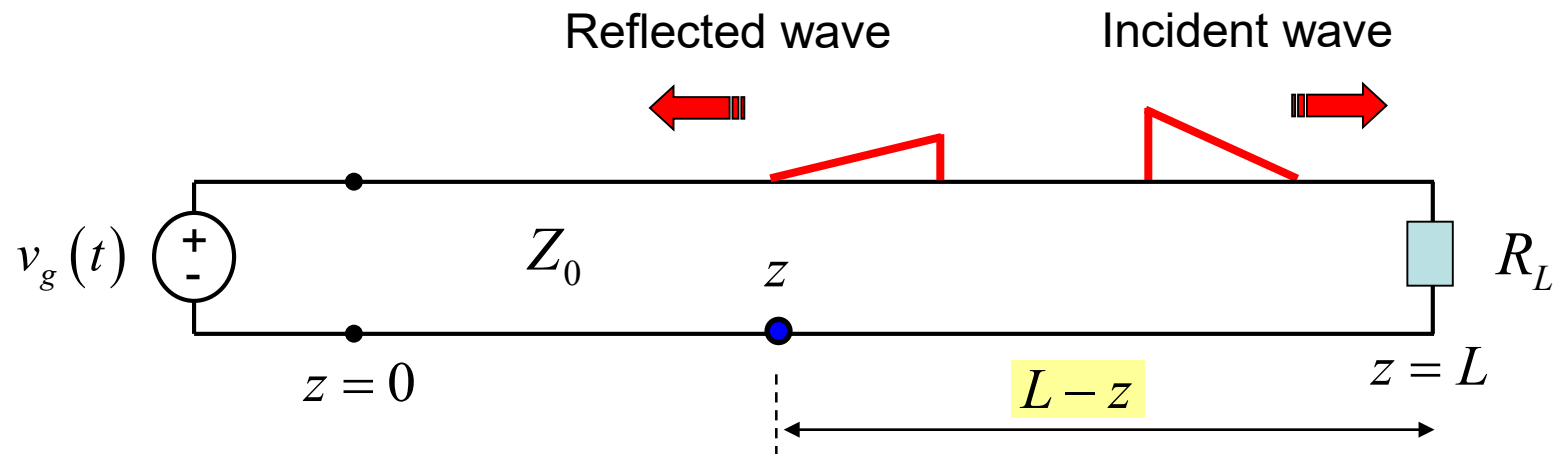
$$-1 \leq \Gamma_L \leq +1$$

Load Reflection (cont.)

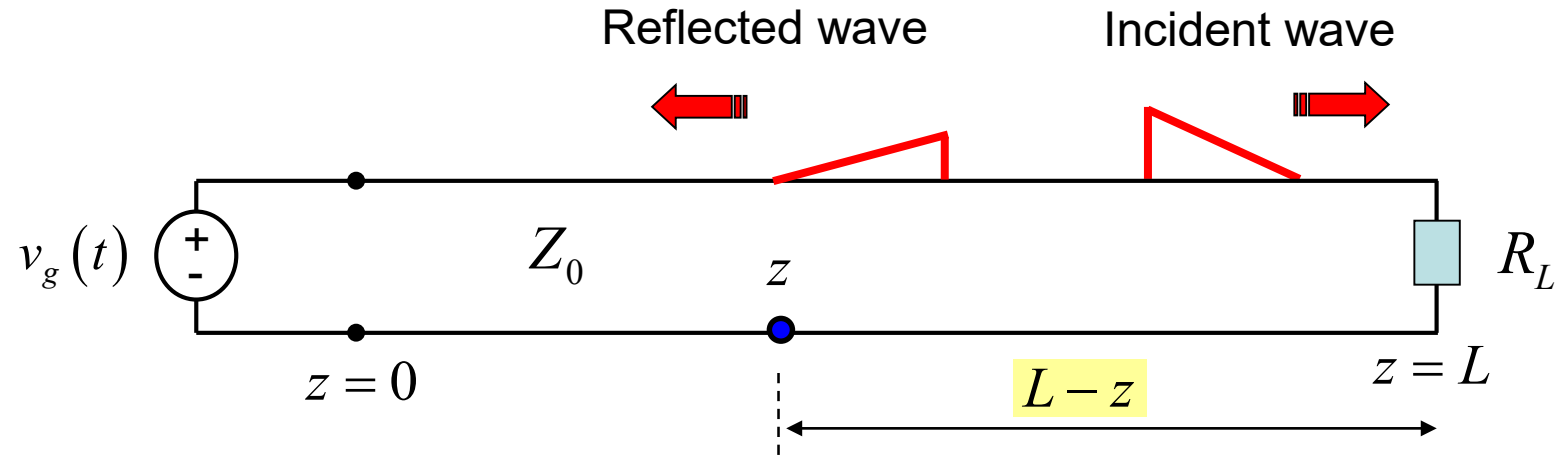
We now proceed to obtain $v^-(z, t)$ (for arbitrary z)

At the load: $v^-(L, t) = \Gamma_L v_L^+(t) = \Gamma_L v_g(t - L / c_d)$

Moving back from the load, we see an additional delay of $(L - z) / c_d$



Load Reflection (cont.)



Hence, we have for the reflected wave:

$$v^-(z, t) = \Gamma_L v_g \left(t - L / c_d - (L - z) / c_d \right)$$

Reflection Picture

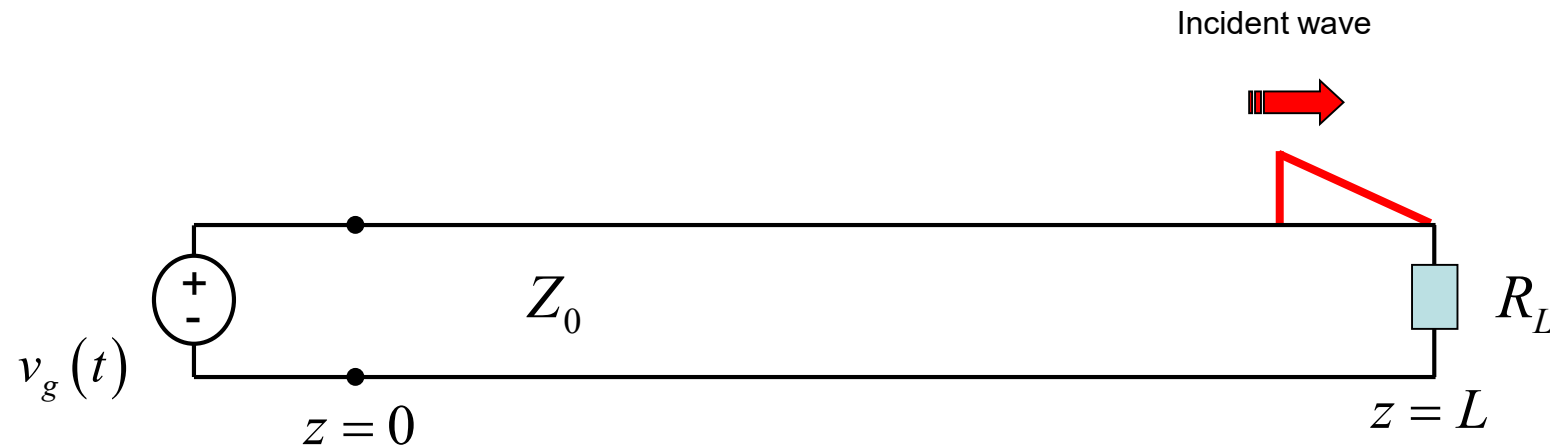
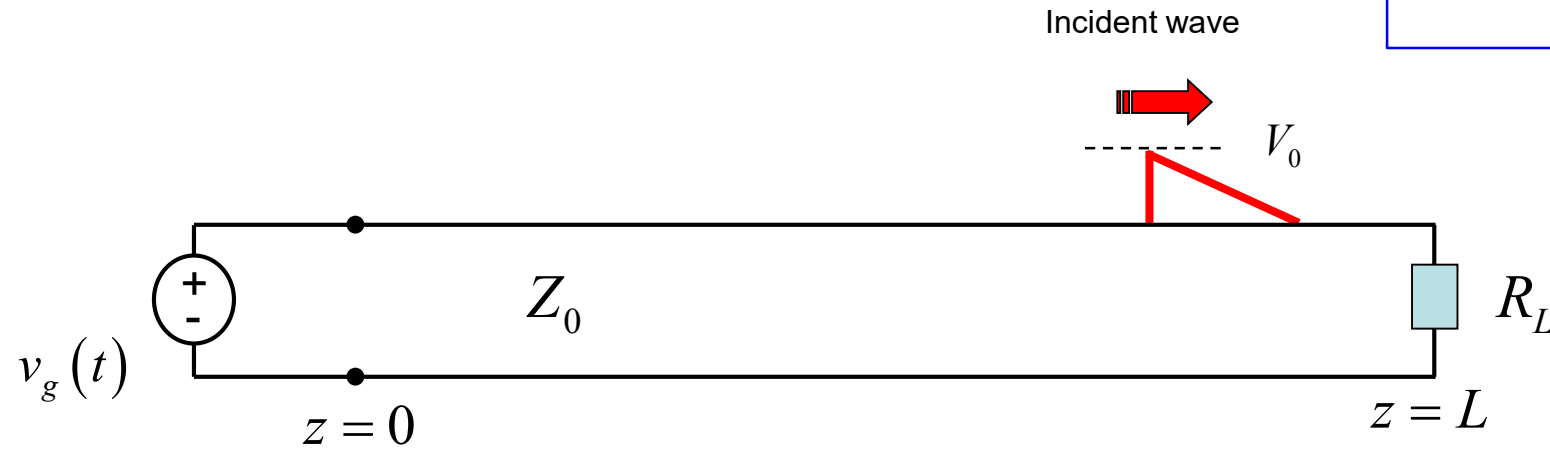
Assume:

$$Z_0 = 50 \, [\Omega]$$
$$R_L = 25 \, [\Omega]$$

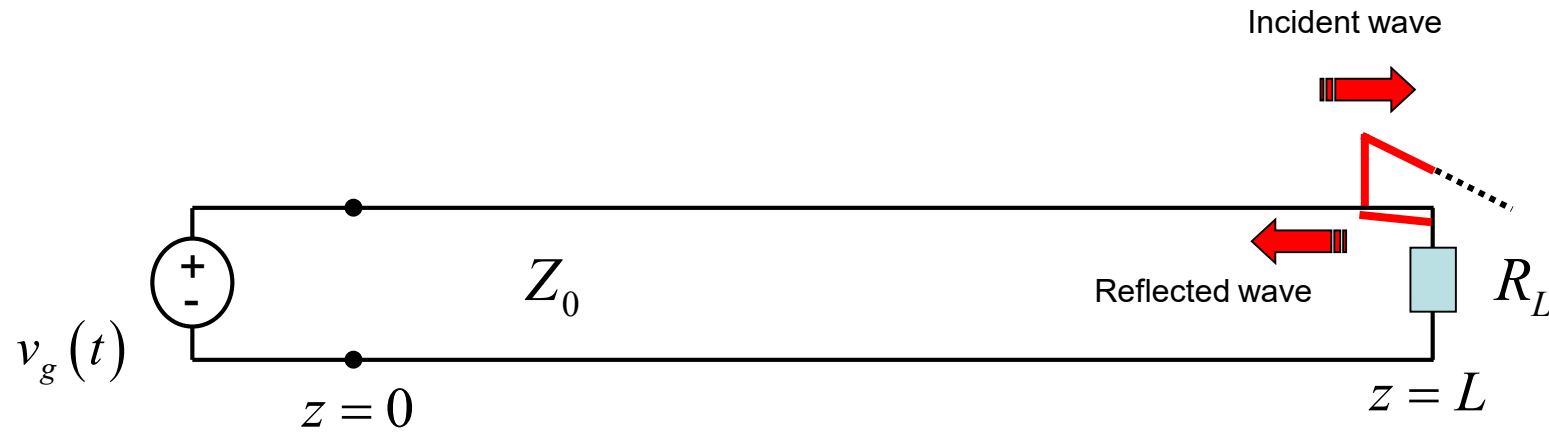
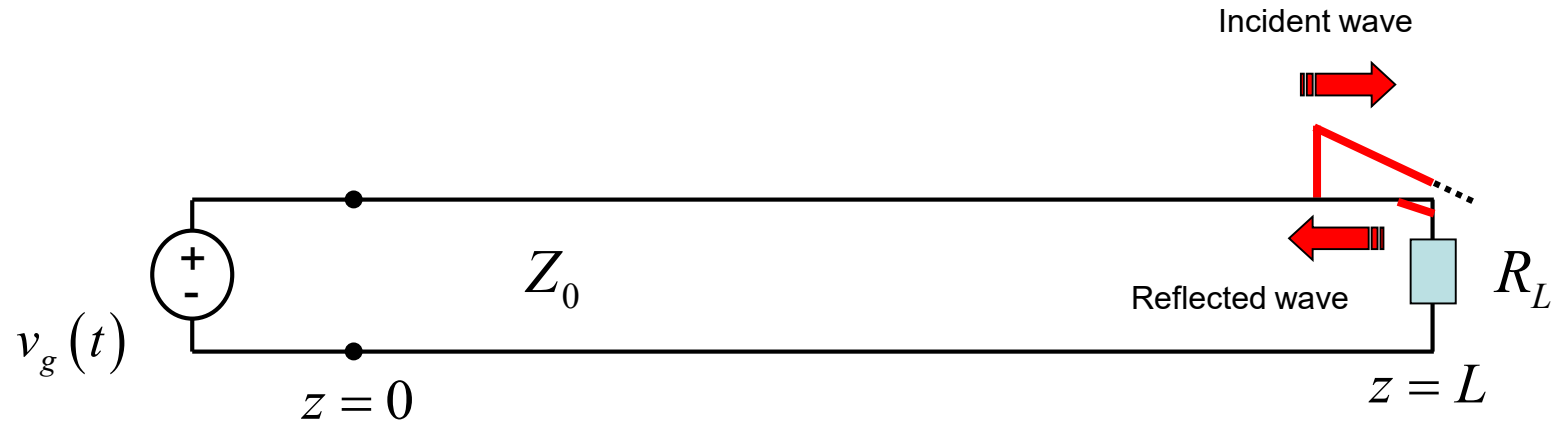


$$\Gamma_L = -1/3$$

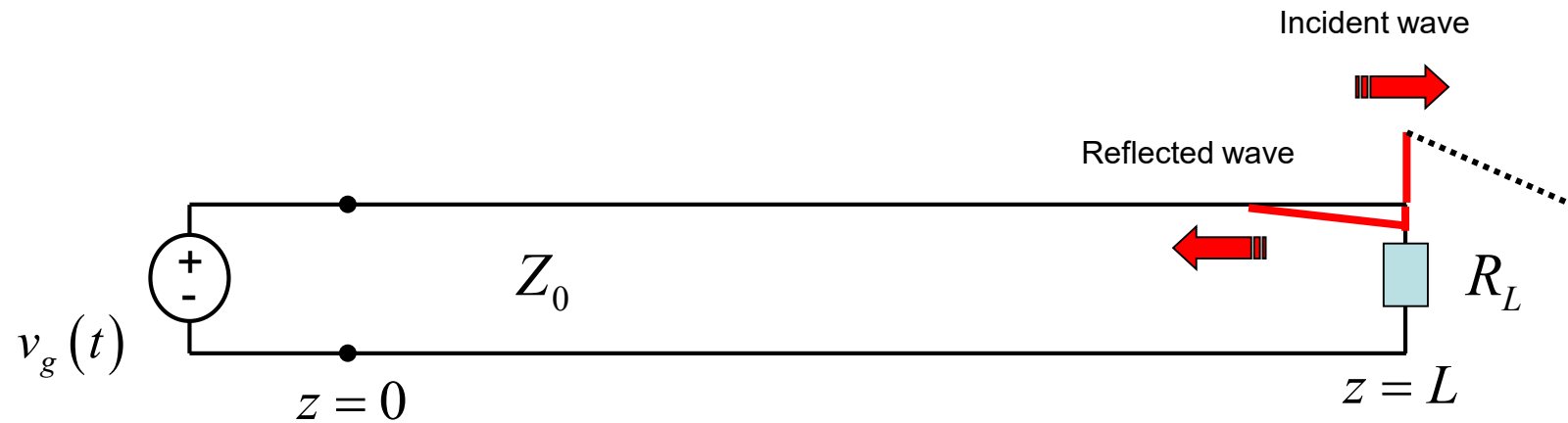
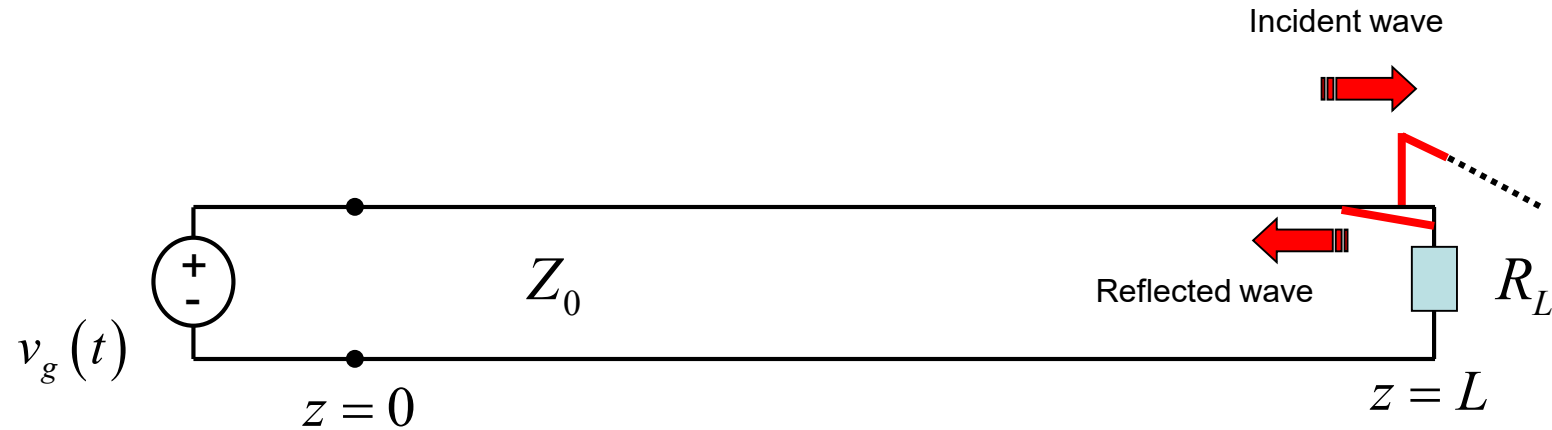
Here we have a series of snapshots, showing how the incident wave reflects from the load.



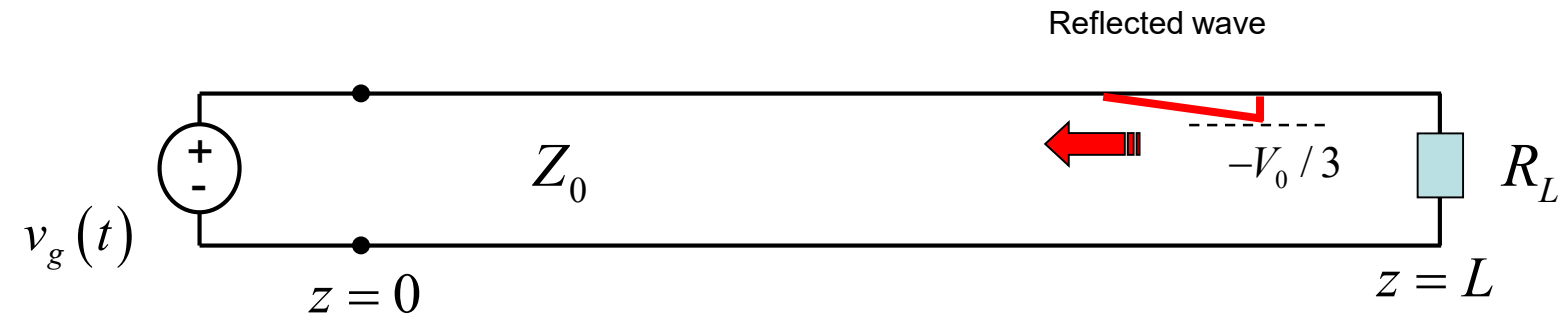
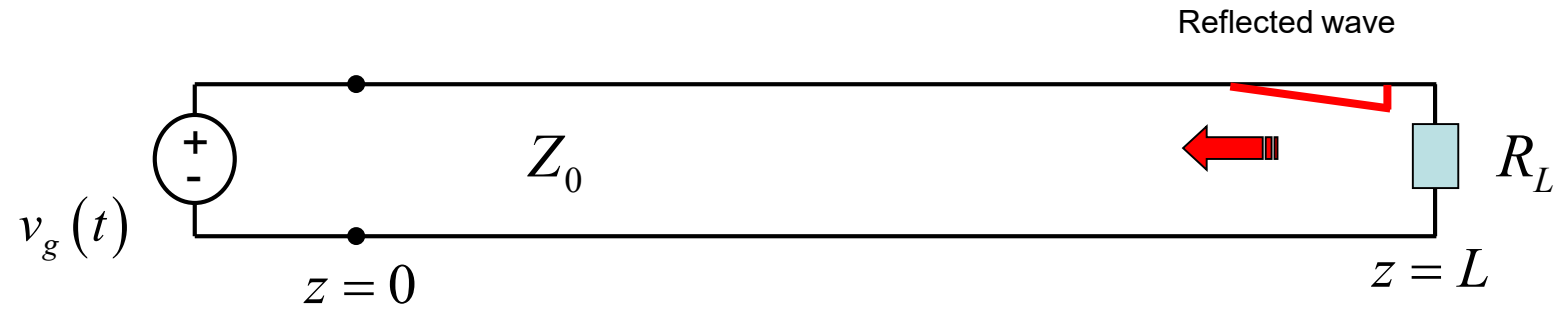
Reflection Picture (cont.)



Reflection Picture (cont.)



Reflection Picture (cont.)



Practical Voltage Generator

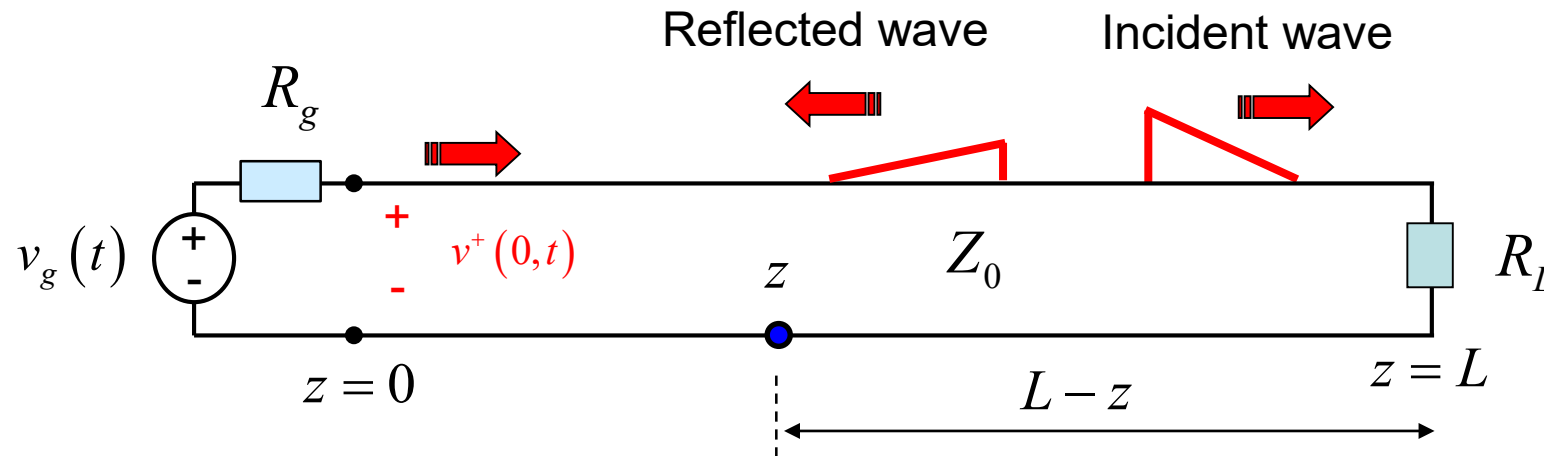
Including Thévenin Resistance of Source

Voltage divider:

$$v^+(0, t) = A v_g(t); \quad A = \frac{Z_0}{R_g + Z_0}$$

Note:

The incident wave does not “see” the load as it starts traveling on the line.



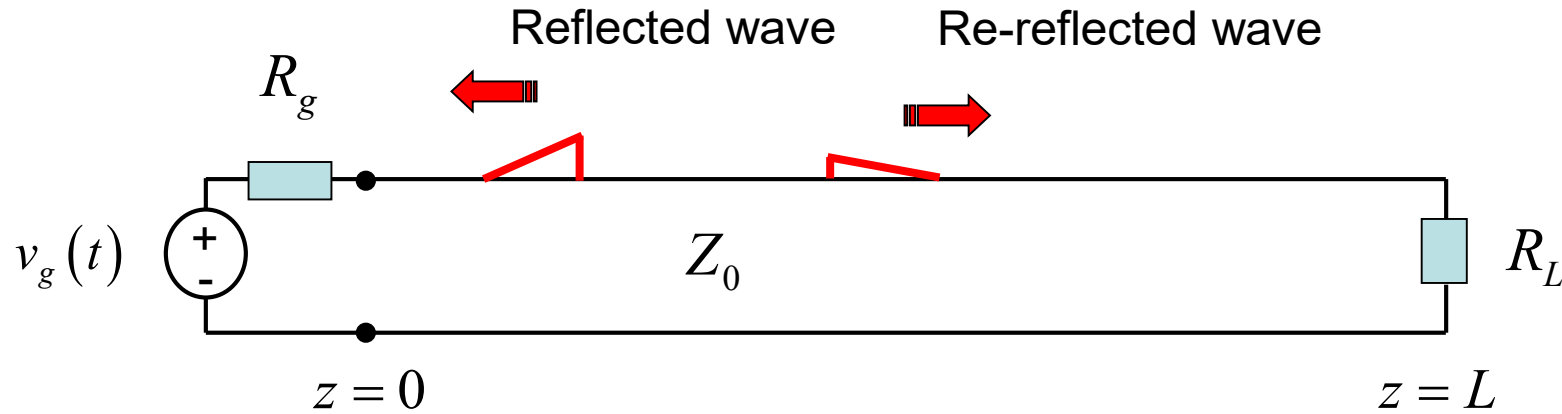
Incident wave: $v^+(z, t) = A v_g(t - z / c_d)$

Reflected wave: $v^-(z, t) = \Gamma_L A v_g(t - L / c_d - (L - z) / c_d)$

Reflections at Source End

After the reflected wave hits the source end, there will be another reflection.

Here we allow the source to have a Thévenin resistance R_g .

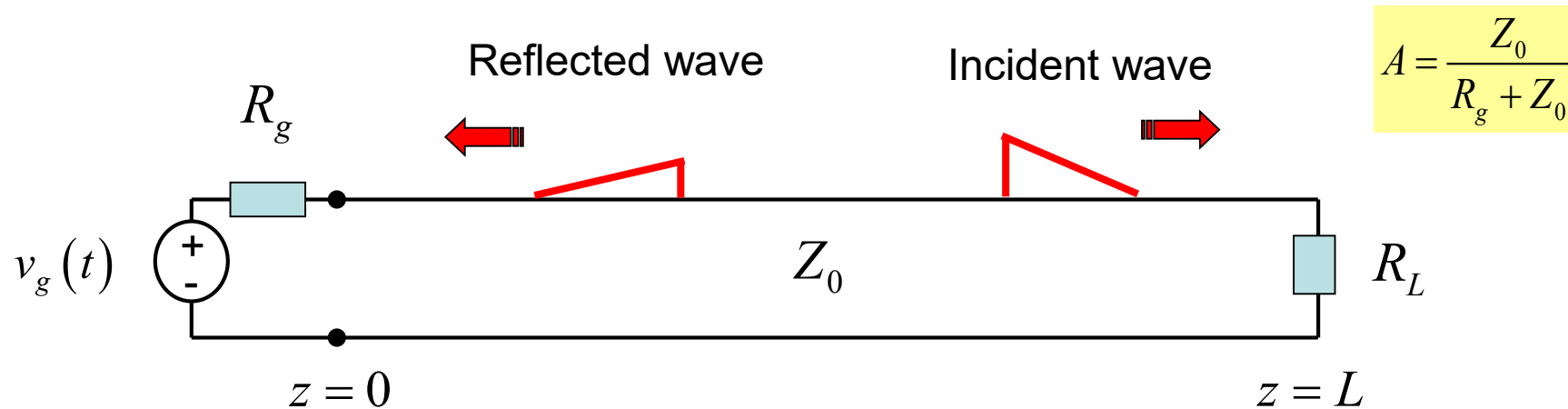


$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right)$$

Note:

In calculating the reflection from the source, we ignore the voltage source.
(We have already accounted for the voltage source in launching the incident wave.)

Complete Wave Picture



Here are the first four waves:

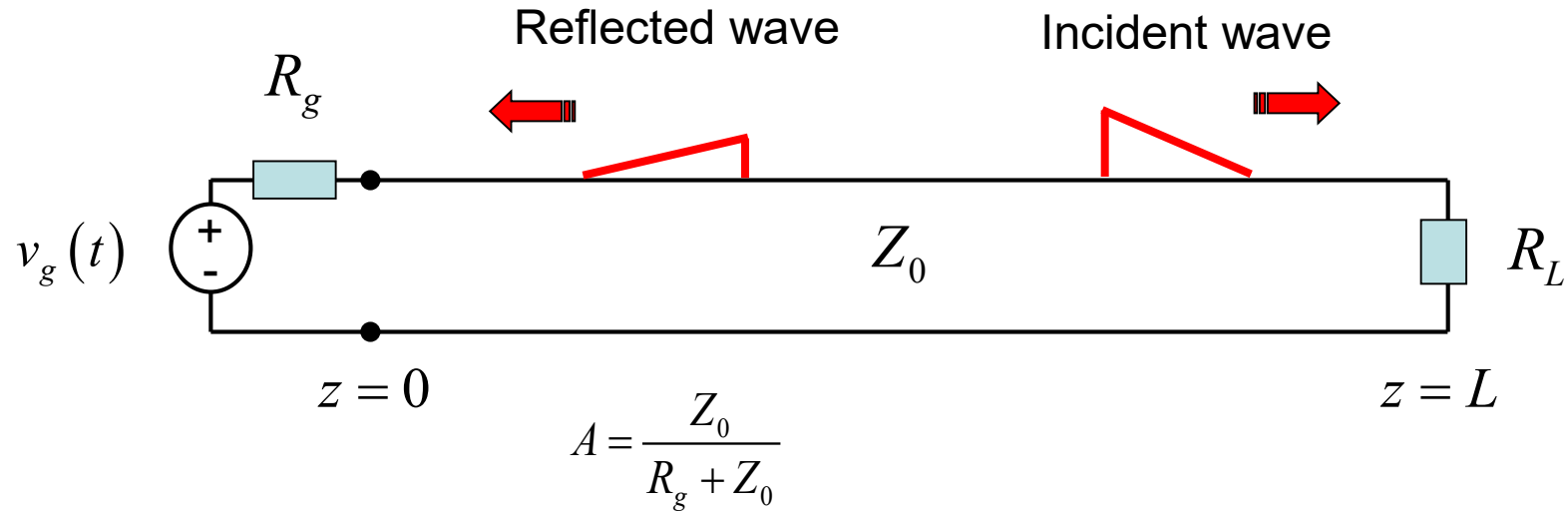
Incident: $v^+(z, t) = A v_g(t - z / c_d)$

Reflected: $v^-(z, t) = \Gamma_L A v_g(t - L / c_d - (L - z) / c_d)$

Re-reflected: $v^{++}(z, t) = \Gamma_g \Gamma_L A v_g(t - 2L / c_d - z / c_d)$

Re-re-reflected: $v^{--}(z, t) = \Gamma_g \Gamma_L^2 A v_g(t - 3L / c_d - (L - z) / c_d)$

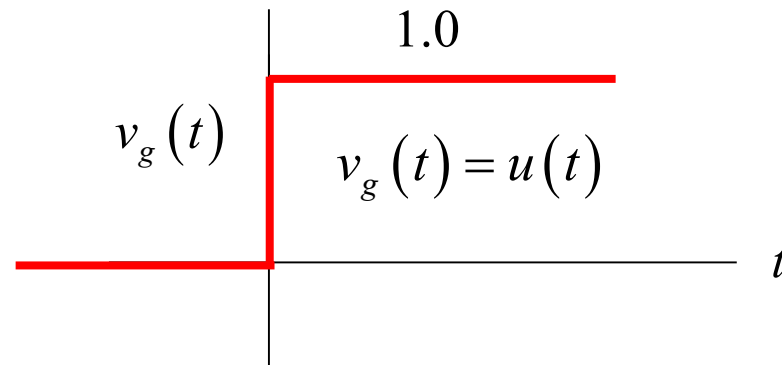
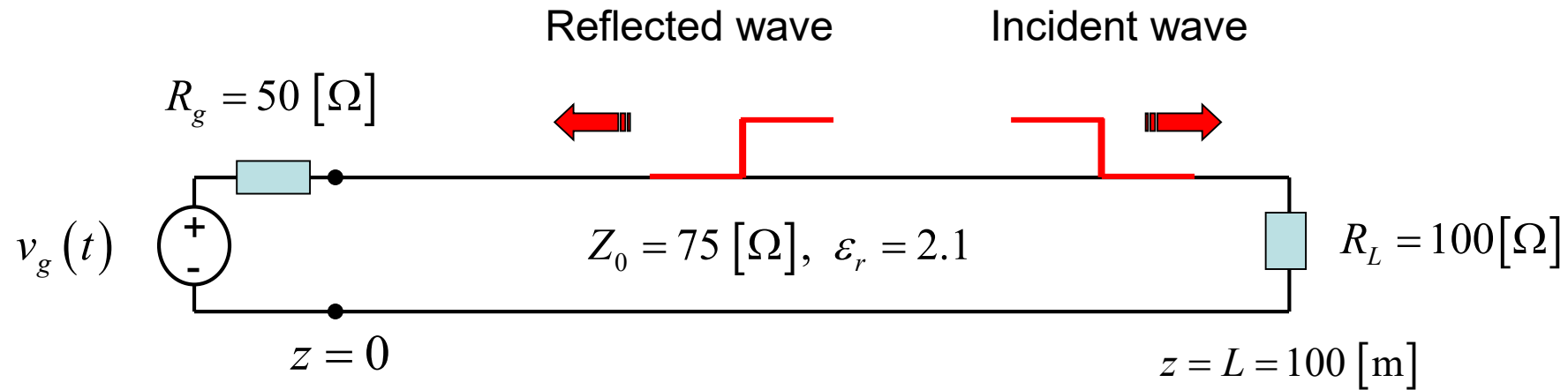
Exact Solution



The exact solution (from summing all the reflections):

$$v(z, t) = A \sum_{\substack{n=0 \\ \text{even}}}^{\infty} (\Gamma_g \Gamma_L)^{n/2} v_g(t - nL / c_d - z / c_d) \\ + A \Gamma_L \sum_{\substack{n=1 \\ \text{odd}}}^{\infty} (\Gamma_g \Gamma_L)^{(n-1)/2} v_g(t - nL / c_d - (L - z) / c_d)$$

Example



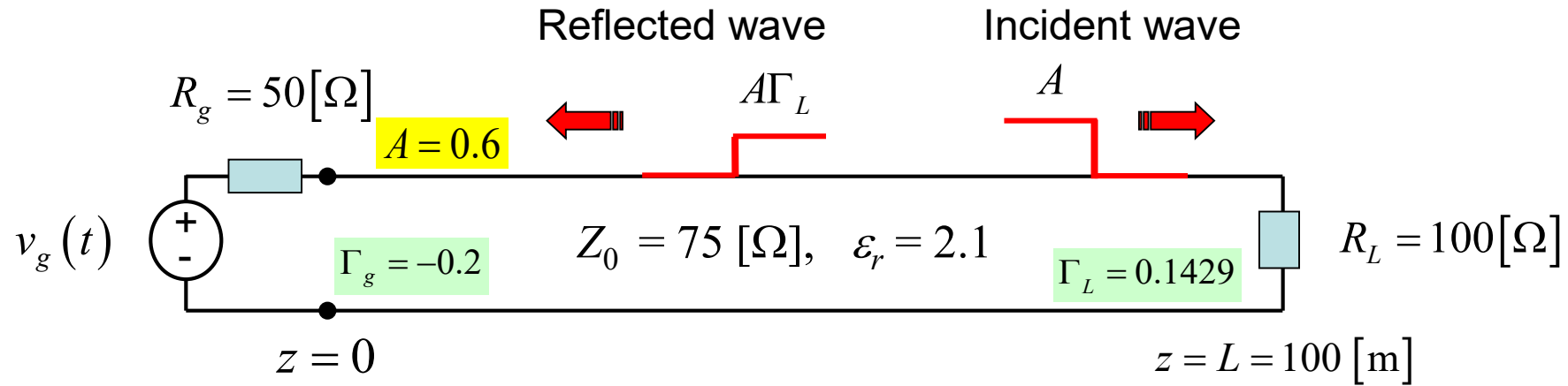
$$c_d = c / \sqrt{2.1} = 2.069 \times 10^8 [\text{m/s}]$$

$$A = \frac{75}{75 + 50} = 0.6000$$

$$\Gamma_L = \frac{100 - 75}{100 + 75} = 0.1429$$

$$\Gamma_g = \frac{50 - 75}{75 + 50} = -0.2000$$

Example (cont.)



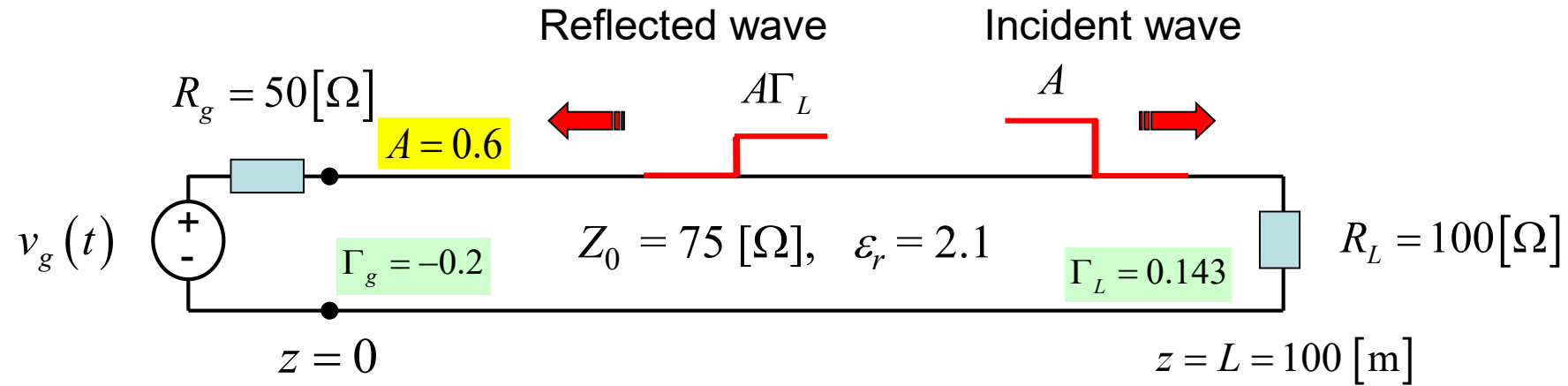
Incident: $v^+(z, t) = (0.6)u(t - z / c_d)$

Reflected: $v^-(z, t) = (0.1429)(0.6)u(t - L / c_d - (L - z) / c_d)$

Re-reflected: $v^{++}(z, t) = (-0.2)(0.1429)(0.6)u(t - 2L / c_d - z / c_d)$

Re-re-reflected: $v^{--}(z, t) = (-0.2)(0.1429)^2(0.6)u(t - 3L / c_d - (L - z) / c_d)$

Example (cont.)



Total voltage:

$$\begin{aligned}
 v(z, t) = & (0.6)u(t - z / c_d) \\
 & + (0.0858)u(t - L / c_d - (L - z) / c_d) \\
 & + (-0.0172)u(t - 2L / c_d - z / c_d) \\
 & + (-0.00245)u(t - 3L / c_d - (L - z) / c_d) \\
 & + \dots
 \end{aligned}$$

$$L = 100 [\text{m}]$$

$$c_d = c / \sqrt{2.1} = 2.069 \times 10^8 [\text{m/s}]$$

Comments

- ❖ The higher-order reflected waves get smaller, due to the reflection coefficients.
- ❖ The bounce diagram (discussed in the next set of notes) gives us a convenient way of tracking all the waves and determining the waveform observed at any point on the line, when the generator voltage is a step function (or a rectangular pulse).

For an arbitrary waveform, we can use the series solution.